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Linux for the Cloud Native Engineer

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What is an operating system?

> “System software that manages computer **hardware** and **software resources**, and provides **common services** for computer programs”
>
> -wikipedia

Why does the OS matter

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  - Namespaces, cgroups, capabilities, …

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- Infrastructure as code -> compute, storage, networking
- Observability and monitoring -> metrics and traces
  - CPU, memory, scheduling, system calls, …

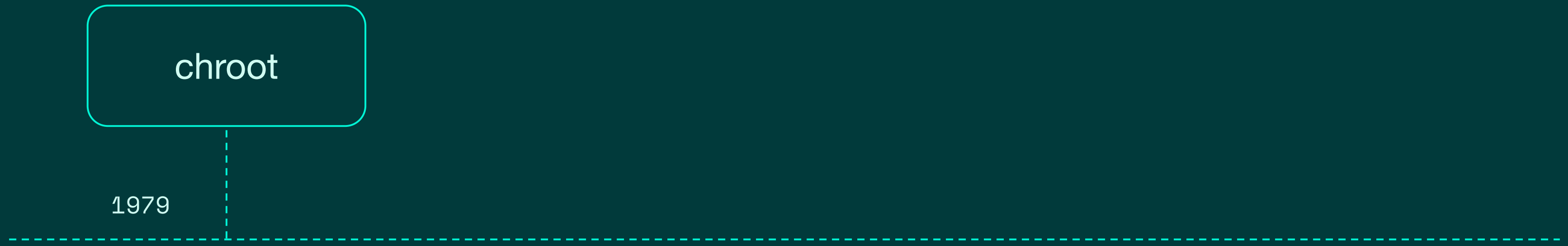
Problems with your container that are actually kernel problems

- Performance bottlenecks
- Container permissions
- Resource utilization

History of Containers: pre-history

- Virtual machines
  - To isolate a process, bring a whole operating system
- Container = OS-level virtualization
  - Virtualization within the operating system

History of Containers



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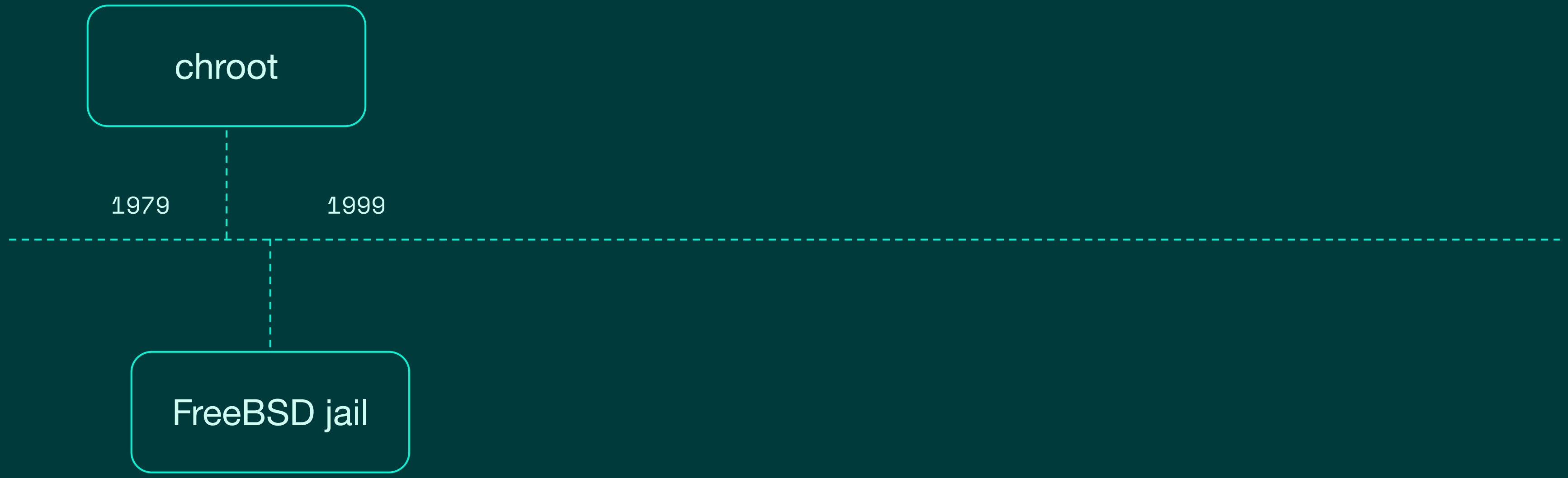

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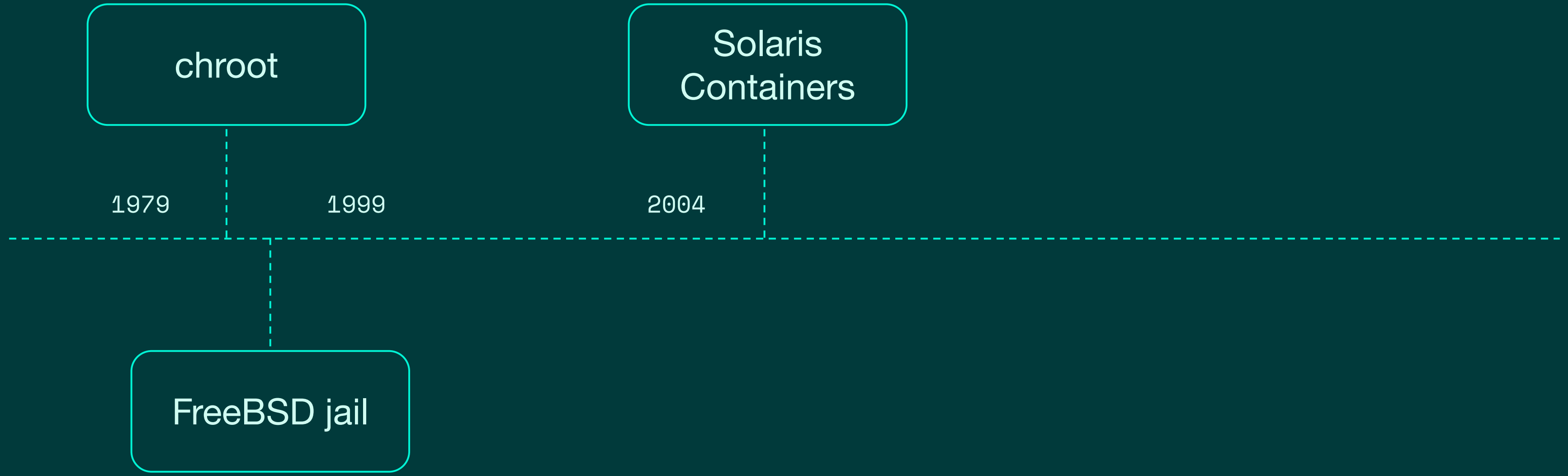

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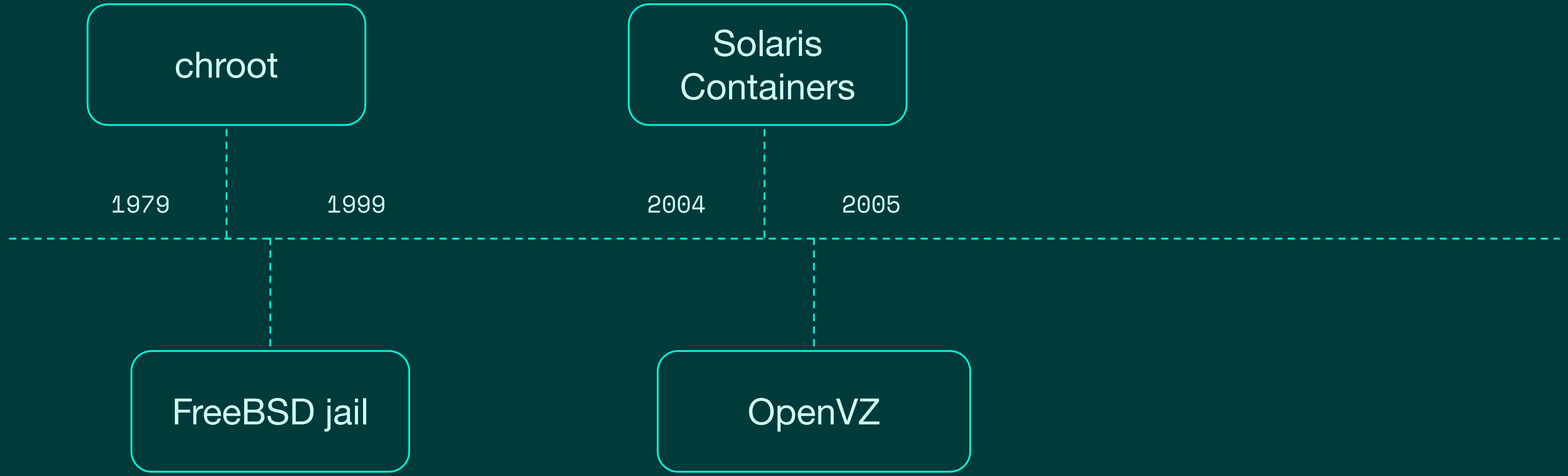

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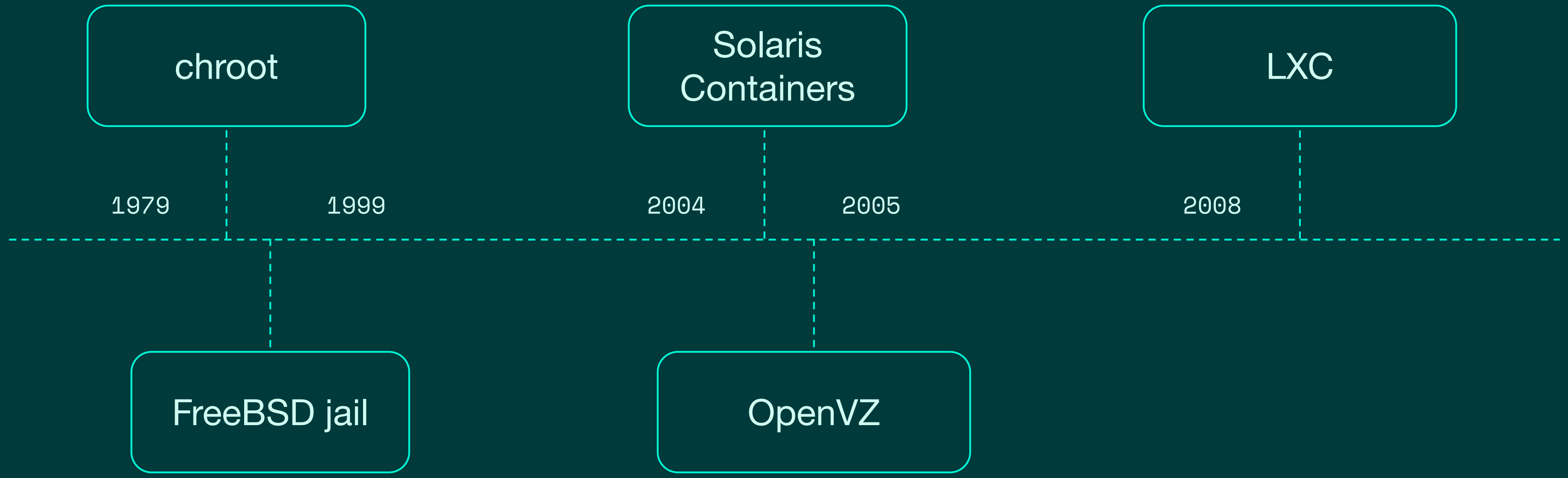

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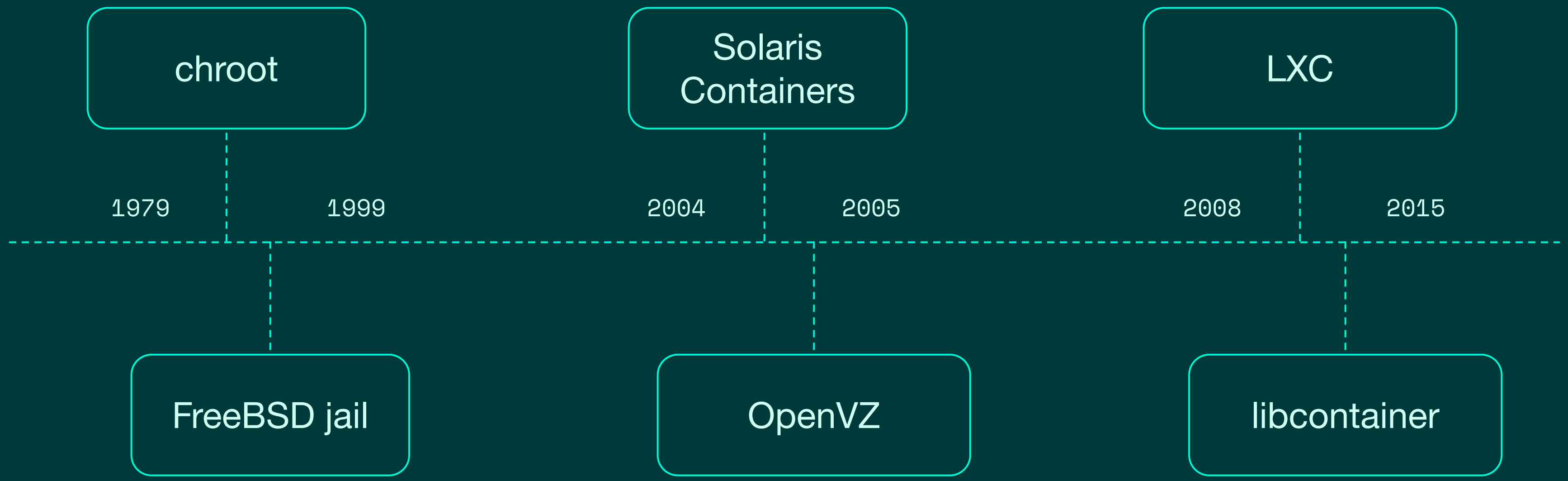
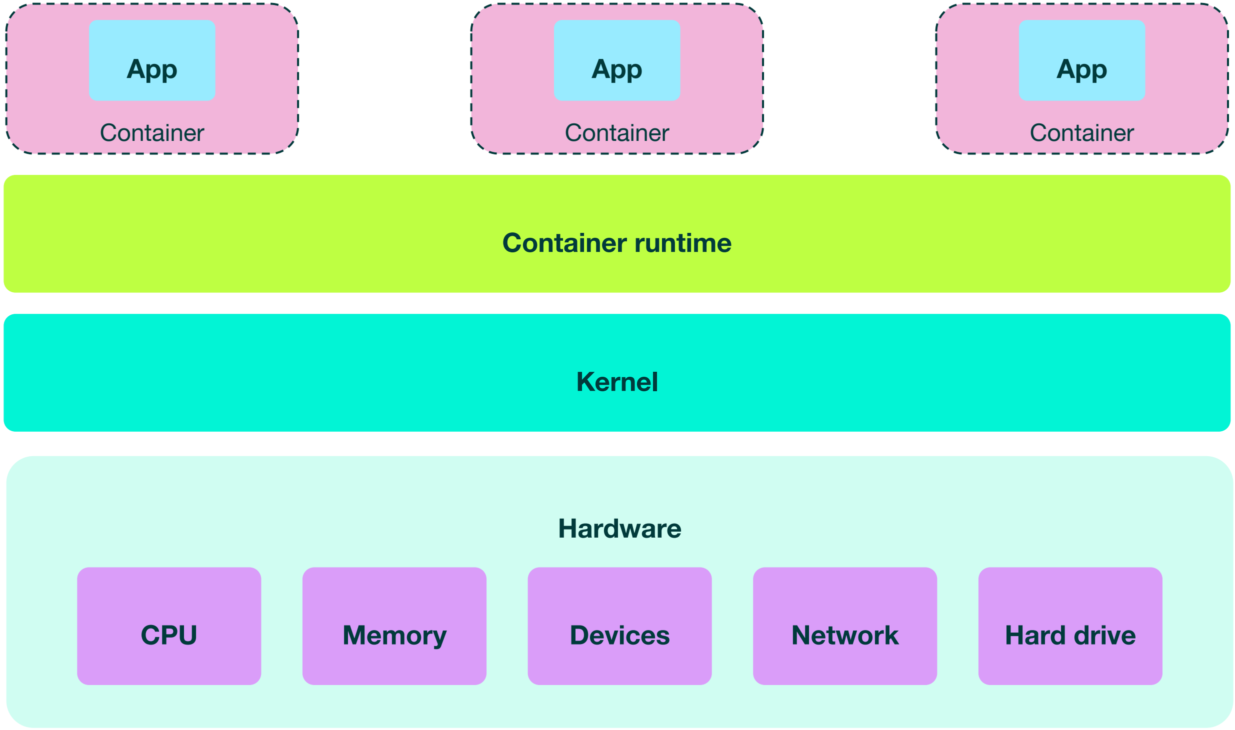


Containers: where we are now

- OS virtualization, but with:
  - Namespaces
  - Cgroups
  - Capabilities
  - security profiles
  - network interfaces
  - …

But underneath…



Devices

- Hardware components the OS interacts with
- Block devices
  - /proc/partitions
  - You can see all the devices, even if you don’t have permission to access them

System calls

- How users request services from the OS
  - Process control
  - Management of files, devices, memory
  - Communication
- System calls run in the OS
  - Even with user namespaces

Processes

- OS creates all processes
- Scheduling between containers
- Memory allocation to process

Resources

- CPU and memory usage
- Can view full usage from container
  - /proc/cpuinfo
  - /proc/meminfo

Summary

- Containers run like processes in the OS kernel
- Namespaces, cgroups, capabilities, etc are all features of Linux
- The OS manages hardware resources
- So understanding the OS help you understand your containers

Thank You

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